NARAYANA COACHING CENTER
NARAYANA IIT-JEE / NEET / FOUNDATIONS
45 YEARS OF EXCELLENCE

Sec: XI_STU_IC_IIT
Date:01-07-2024
Time: 3 Hrs.
FTM – 03
Max.Marks:300

JEE MAIN 2022 Model

PHYSICS

Section	Question Type	+Ve Marks	–Ve Marks	No.of Qs.	Total marks
Sec – I (Q.N : 1 – 20)	Questions with Single Correct Choice	4	–1	20	80
Sec–II (Q.N : 21 – 30)	Questions with Integer Answer Type(+ / –Number)	4	–1	10	20
Total				30	100

CHEMISTRY

Section	Question Type	+Ve Marks	–Ve Marks	No.of Qs.	Total marks
Sec – I (Q.N : 31 – 50)	Questions with Single Correct Choice	4	–1	20	80
Sec–II (Q.N : 51 – 60)	Questions with Integer Answer Type (+ / – Number)	4	–1	10	20
Total				30	100

MATHEMATICS

Section	Question Type	+Ve Marks	–Ve Marks	No.of Qs.	Total marks
Sec–I (Q.N : 61 – 80)	Questions with Single Correct Choice	4	–1	20	80
Sec–II (Q.N : 81 – 90)	Questions with Integer Answer Type (+ / – Number)	4	–1	10	20
Total				30	100

Exam Syllabus

PHYSICS	**40% :** Motion under gravity, Graphs **40% :** KINEMATICS: Uniformly accelerated motion (1D) **20% :** MATHEMATICAL PHYSICS: Application of differentiation in physics, Maxima & Minima Integration and its application in physics Unit and Dimension: Introduction of physics, physical quantities and physical laws Fundamental and derived quantities and ways to write dimension, Unit and Dimension Uses of dimensional analysis (Principle of homogeneity) Unit and Dimension: Significant figures, Errors in measurements Problems discussion from material KINEMATICS: Frame of Reference, Mathematical Expression and definition of position, distance and displacement Avg velocity& instantaneous velocity, Avg speed & instantaneous speed, acceleration
CHEMISTRY	**40% :** Equivalent weight and related numericals (excluding redox) Normality and equivalent concept, **40% :** Percentage calculation of elements, Formula representation of molecule: (i) Empirical formula, (ii) Molecular formula, (iii) Formula unitn-factor of acid, base and salts **20% :** Nature of Matter, Properties of matter & their measurement SI unit, Mass weight, Volume temperature, Need for standard reference, Scientific notation, Precision & accuracy, Significant figure and calculation involving significant figure. Law of chemical combination : (i) Law of conservation of mass, (ii) Law of constant combination, (iii) Law of multiple proportion, (v) Gay Lussac's law, (vi) Avagdro law, (vii) Dalton's atomic theory Atomic mass Mole concept: Calculation based on mole concept, concept of gram atom, gram molecule Percentage by mass (w/w), percentage by volume (v/v) Concentration terms: Molarity, molality, mole fraction and their interconversions, relation between density & molarity Concentration terms and their interconversions, relation between density & molarity Question practice on concentration Concept of limiting reagent : Use of stoichiometry in balanced equation & limiting reagent concept
MATHEMATICS	**40% :** maximum and minimum value of trigonometric expressions, analysis of the form y=asinx+bcosx. conditional identities, sum of Trigonometric series. **40% :** Domain and ranges of trigonometric function, Graph of trigonometric functions. Trigonometric ratio of multiple angles, sub-multiple angles. **20% :** SEQUENCE AND SERIES Introduction of Sequence and Series, nth term of AP. Properties of A.P. Sum of n term of an AP, Arithmetic Mean, nth term of G.P., sum of n term of GP. Properties of G.P., Geometric mean, A.G.P., Introduction to HP, HM, Relation between AM, GM and HM Difference method, Vn method, special series, TRIGONOMETRIC FUNCTIONS Angle, system of measurement of angles, trigonometric function, Values of trigonometric function for some specific angles, Trigonometric ratios of allied angles, Transforming product into sum or difference, Transforming the sum or difference into product.

Rough Work

PHYSICS **Max.Marks:100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

1. A particle is projected vertically upward with velocity 25 m/s. Displacement (m) during 3rd second of motion. (Take g = 10 m/s^2)
   A) 2 B) 3
   C) 4 D) 0
2. Two objects A and B are thrown upward simultaneously with same speed. The mass of A is greater than that of B. Suppose the air exerted a constant and equal force of resistance on the two bodies.
   A) The two bodies will reach the same height
   B) A will go higher than B
   C) B will go higher than A
   D) None of these
3. A person throws vertically up n balls per second with same velocity. He throws a ball whenever the previous one is at its highest point. The height to which the balls rise is
   A) $\frac{g}{n^2}$ B) $2gn$
   C) $\frac{g}{2n^2}$ D) $2gn^2$
4. A particle is projected vertically upwards from ground at time $t = 0$; The ball is observed at same height at time t_1 and t_2. The maximum height attained by particle is ______
   A) $\frac{1}{2}g(t_1+t_2)^2$ B) $\frac{1}{8}g(t_1+t_2)^2$
   C) $\frac{1}{2}g\,t_1t_2$ D) $g\,t_1t_2$
5. A ball is released from the top of a building 180 m high it takes time t to reach the ground with what speed should it be projected vertically downward so that it reaches the ground in time $\frac{5t}{6}$.
   A) 10 m/s B) 12 m/s
   C) 11 m/s D) 14 m/s
6. The distance covered by a freely falling body in its first, second, third............nth seconds of its motion.
   A) Forms an Arithmetic progression
   B) Forms a Geometric progression
   C) Do not form any well defined series
   D) Forms Arithmetic and Geometric progression both

Rough Work

7. A falling stone takes 0.2 s to fall vertically across a window which has length 1 m. From how high above the top of the window was the stone dropped. (Take g = 10 m/s^2)
   A) 0.8 m B) 1 m
   C) 2.8 m D) 1.8 m

8. A stone is projected in vertically upward direction with speed 5 m/s from a building of height *H*. Stone takes 5 sec to reach the ground. The value of *H* is.............(Take g = 10 m/s^2)
   A) 25 m B) 30 m
   C) 100 m D) 50 m

9. A body starts moving with velocity V_1 at $t = 0$. The body is accelerated uniformly. At time $t = t_1$ it has velocity V_2. Assume body is moving in straight line. Average velocity of body is -
   A) $\frac{V_1 - V_2}{2}$ B) $\frac{2V_1 - V_2}{3}$
   C) $\frac{V_1 + V_2}{2}$ D) $\frac{V_2 - V_1}{2}$

10. A particle is moving with speed $v = b\sqrt{x}$ along positive *x*-axis. Calculate the speed of the particle at time $t = \tau$ (assume that the particle is at origin at $t = 0$).
   A) $\frac{b^2\tau}{4}$ B) $\frac{b^2\tau}{2}$
   C) $b^2\tau$ D) $\frac{b^2\tau}{\sqrt{2}}$

11. A person travels x distance with velocity v_1 and then x distance with velocity v_2 in the same direction. The average velocity of the person is v, then the relation between v, v_1 and v_2 will be:
   A) $v = v_1 + v_2$ B) $v = \frac{v_1 + v_2}{2}$
   C) $\frac{2}{v} = \frac{1}{v_1} + \frac{1}{v_2}$ D) $\frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}$

12. A scooter accelerates from rest for time t_1 at constant rate a_1 and then retards at constant rate a_2 for time t_2 and comes to rest. The correct value of $\frac{t_1}{t_2}$ will be
   A) $\frac{a_1 + a_2}{a_1}$ B) $\frac{a_1}{a_2}$
   C) $\frac{a_2}{a_1}$ D) $\frac{a_1 + a_2}{a_2}$

13. A particle is moving along x-axis. Its velocity is given as $v = 3t^2 + 4t$. Where *v* is measured in (m/s) *t* is measured in second. Displacement of particle between $t = 0$ to 1 sec.
   A) 4m B) 3m
   C) 2m D) 1m

Rough Work

14. A particle starts moving from rest with constant acceleration a. Ratio of distances travelled in first 4 sec. and in next 4 sec.

A) $\frac{1}{2}$
B) 1
C) $\frac{1}{3}$
D) $\frac{1}{4}$

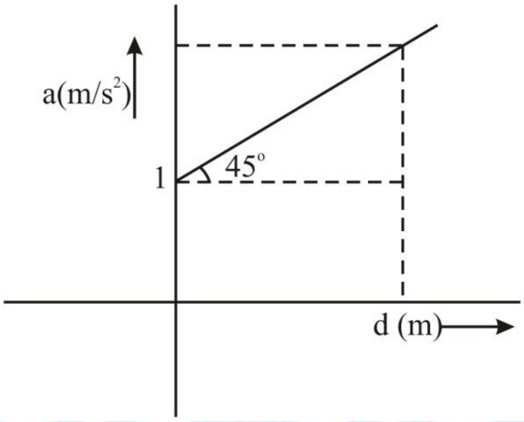
15. A particle is moving with velocity 10 m/s along positive x-axis. At $t=0$, acceleration of 2 m/s^2 is applied on particle along negative x-axis. The time at which particle is speed become zero, momentarily.

A) 6 s
B) 5 s
C) 10 s
D) 1 sec

16. A man is driving a car with speed 10 m/s. At some distance, he sees a huge pit on the road. After seeing the pit, he applies brakes to stop the car to decelerate the car with 2 m/s^2. Distance travelled by car after seeing the pit is S. The value of S, if reaction time of man is 0.5 sec.

A) 30 m
B) 25 m
C) 20 m
D) 50 m

17. Acceleration versus displacement graph of a particle is given below



Initial velocity of particle is zero.
Velocity of particle when acceleration of particle is 3 m/s^2, is -

A) 3 m/s
B) $2\sqrt{2}$ m/s
C) 2 m/s
D) 4 m/s

18. If $y=5+10x-5x^2$, then maximum value of y is -

A) 11
B) 12
C) 5
D) 10

19. $V=t^4+5t^3-6t^2+5$. Find $\frac{dV}{dt}$ at $t=1$.

A) 12
B) 13
C) 7
D) 15

Rough Work

20. Force $F = X + Yt^2$ is given where X and Y are constants Dimensional formula of $\frac{X}{Y}$ is -

A) T^2 B) T^{-2} C) MLT D) T^3

SECTION-II
(Integer Value Answer Type)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals**. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

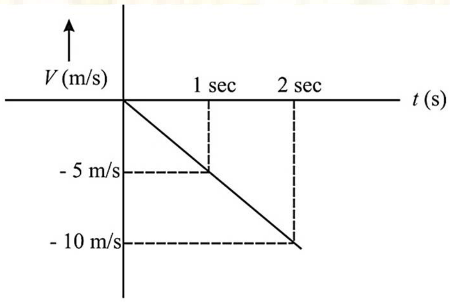
21. A ball is dropped from a building of height *H*. The ratio of the times taken to cover the first one-fourth of *H* and last one-fourth of *H* – is given as $a+\sqrt{b}$. Then value of $a+b$ is...............

22. The position of an object moving along the *x*-axis is given by $x = a + bt^2$ where $a = 8.5\,\text{m}$ and $b = 2.5\,\text{m/s}^2$ and t is measured in seconds. The average velocity of the object between $t = 1\,\text{sec}$ and $t = 3\,\text{sec}$ is V. Then value of $\frac{V}{5}$ is................

23. A particle is moving along the x-axis the position *x* at time *t* is given by $t = \sqrt{x} + 4$, where *x* is in metres and *t* is in seconds. What is the speed of the particles at $t = 2\,\text{sec}$............

24. Force $F = (10 \pm 0.1)\,N$ is applied perpendicularly on a rectangular sheet of length $(20 \pm 0.2)\,\text{mm}$ and breadth $(10 \pm 0.2)\,\text{mm}$. The maximum percentage error in calculating pressure...............

25. On a planet named Titan acceleration due to gravity is *g*. A ball is dropped from height *H* above the ground of the planet. The velocity versus time graph is given below –
What will be value of g (m/s^2)?



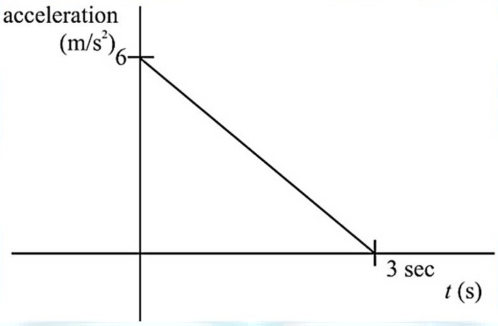
Rough Work

26. Position of a particle is given as $x = 10t - t^2$. The distance covered by particle from $t = 0$ to $t = 7$ sec. is S. Then value of $\left(\frac{S+1}{5}\right)$ is.................

27. If Force F, Velocity V and Time T are considered as fundamental physical quantity, the dimensional formula of density is given as $F^x V^{-4} T^y$. Then value of $|x+y|$ is ....

28. The velocity of the bullet becomes half after travelling 3 cm in a wooden block. Assuming that bullet is facing a constant resistance during its motion in the block. The bullet stops completely after travelling further distance y (cm) the value of y is ....

29. A body stars from rest at $t = 0$ the acceleration – time graph is shown in the figure. The maximum velocity attained by the body will be....................



30. A ball is projected vertically upward with initial velocity of 40 m/s at $t = 0$ s. At $t = 2$ s, another ball is projected vertically upward with same velocity. At t = ... s, second ball will meet the first ball. (g = 10 m/s^2)

Rough Work

CHEMISTRY **Max.Marks:100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. How much volume of CO_2 at S.T.P is liberated by the combustion of 100 $cm^3$of propane (C_3H_8)?
A) 100 cm^3 B) 200 cm^3
C) 300 cm^3 D) 400 cm^3

32. A compound of magnesium contains 21.9% magnesium, 27.8% phosphorus and 50.3% oxygen. What will be the simplest formula of the compound?
A) $Mg_2P_2O_7$ B) $MgPO_3$
C) $Mg_2P_2O_2$ D) MgP_2O_4

33. How much weight of H_2SO_4 is required to completely react with 20 g of NaOH.
A) 0.136 g B) 2.72 g
C) 204.5 g D) 24.5 g

34. The equivalent mass of H_3PO_4 in the reaction given below is
$$H_3PO_4 + NaOH \rightarrow NaH_2PO_4 + H_2O$$
A) 49 B) 98
C) 32.6 D) 40

35. The equivalent weight of divalent metal is 31.82. The weight of a single atom is
A) $32.55 \times 6.02 \times 10^{23}$ g B) $32.77 \times 6.02 \times 10^{23}$ g
C) 63.64 g D) $63.64 / 6.02 \times 10^{23}$ g

36. Eq. wt. of metal oxide is 32 then what will eq. wt. of metal.
A) 32 B) 8
C) 24 D) 20

37. Valency Factor of H_3PO_2 is -
A) 1 B) 4
C) 3 D) 2

38. Equivalent weight of ferrous ion is (At. Wt. of Fe = 56)
A) 56 B) 28
C) 14 D) 32

Rough Work

39. Match the column I with column II.

	Column-I (Physical quantity)		Column-II (Unit)
(a)	Molarity	(i)	mol kg^{-1}
(b)	Mole fraction	(ii)	mol
(c)	Mole	(iii)	Pascal
(d)	Molality	(iv)	Unitless
(e)	Pressure	(v)	mol L^{-1}

A) (a) – v, (b) – iv, (c)-ii, (d)-i, (e)-iii
B) (a) – i, (b) – ii, (c)-iii, (d)-iv, (e)-v
C) (a) – ii, (b) – iii, (c)-i, (d)- v, (e)-iv
D) (a) – i, (b) – iii, (c)-v, (d)-iv, (e)-i

40. Which of the following has the highest normality? (Consider each of the acid is 100% ionized.)

A) 1 M H_2SO_4
B) 1 M H_3PO_3
C) 1 M H_3PO_4
D) 1 M HNO_3

41. The density (in g mL^{-1}) of a 3.60 M sulphuric acid solution that is 29% H_2SO_4 (molar mass = 98 g mol^{-1}) by mass will be

A) 1.64
B) 1.88
C) 1.22
D) 1.45

42. In a compound *C*, *H* and *N* atoms are present in 9 : 1 : 3.5 by weight. Molecular weight of compound is 108. Molecular formula of compound is

A) $C_2H_6N_2$
B) C_3H_4N
C) $C_6H_8N_2$
D) $C_9H_{12}N_3$

43. The simplest formula of a compound containing 50% of element *X* (atomic mass 10) and 50% of element Y (atomic mass 20) is

A) XY
B) X_2Y
C) XY_3
D) X_2Y_3

Rough Work

44. Match the column I with column II.

List-I		List-II	
(a)	16 g of $CH_4(g)$	(i)	Weighs 28g
(b)	1g of $H_2(g)$	(ii)	60.2×10^{23} electrons
(c)	1 mole of $N_2(g)$	(iii)	Weighs 32g
(d)	0.5 mol of $SO_2(g)$	(iv)	Occupies 11.4 L volume at STP

Choose the correct answer from the options given below:
A) (a) – i, (b) – iii, (c) – ii, (d) – iv
B) (a) – ii, (b) – iii, (c) – iv, (d) – i
C) (a) – ii, (b) – iv, (c)-iii, (d) – i
D) (a) – ii, (b) – iv, (c) – i, (d) – iii

45. The normality of 0.3 M phosphorus acid (H_3PO_3) is,
A) 0.1
B) 0.9
C) 0.3
D) 0.6

46. In which mode of expression, the concentration of a solution remains independent of temperature?
A) Molarity
B) Normality
C) Formality
D) Molality

47. Which has maximum number of molecules?
A) 7 g N_2
B) 2 g H_2
C) 16 g NO_2
D) 16 g O_2

48. The empirical formula of an acid is CH_2O_2, the probable molecular formula of acid may be:
A) $C_3H_6O_4$
B) CH_2O
C) CH_2O_2
D) $C_2H_4O_2$

49. 9.8 g of H_2SO_4 is present 2 litres of a solution. The molarity of the solution is
A) 0.1 M
B) 0.05 M
C) 0.2 M
D) 0.01 M

50. Which of the following option represents correct limiting reagents in reactions (i), (ii) and (iii) respectively?

(i) $\underset{(26g)}{C} + \underset{(20g)}{O_2} \rightarrow CO_2$

(ii) $\underset{(60g)}{N_2} + \underset{(80g)}{3H_2} \rightarrow 2NH_3$

(iii) $\underset{(100g)}{P_4} + \underset{(200g)}{3H_2} \rightarrow P_4O_6$

A) C, N_2, O_2
B) C, N_2, P_4
C) O_2, H_2, P_4
D) O_2, N_2, P_4

Rough Work

SECTION-II
(Integer Value Answer Type)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals**. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. The equivalent mass of the metal if 10 gm of metal form 12 gm of it's oxide on heating in air ....

52. *n* – factor of Potash Alum is ....

53. What is the sum of the n-factor of H_3PO_2, H_3PO_3 & H_3PO_4?

54. The Total no. of electrons in one molecule of C_3O_2 is ....

55. Number of hydrogen atoms per molecule of a hydrocarbon A having 85.8% carbon is ....
(Given : Molar mass of A = 84 g mol^{-1})

56. Chlorophyll extracted from the crushed green leaves was dissolved in water to make 2L solution of Mg of concentration 48 ppm. The number of atoms of Mg in this solution is $x \times 10^{20}$ atoms. The value of x is .... (Nearest Integer)
(Given : Atomic mass of Mg is 24 g mol^{-1}; $N_A = 6.02 \times 10^{23}$ mol^{-1})

57. The complete combustion of 0.492 g of an organic compound containing 'C', 'H' and 'O' gives 0.793g of CO_2 and 0.442 g of H_2O. The percentage of oxygen composition in the organic compound is .... (Nearest Integer)

58. The number of atoms in 8 g of sodium is $x \times 10^{23}$. The value of x is .... (Nearest integer)
(Given : $N_A = 6.02 \times 10^{23}$ mol^{-1}, Atomic mass of Na = 23.0 u)

59. The number of significant figures in 0.00340 is ....

60. The weight of 1×10^{22} molecules of $CuSO_4.5H_2O$ is .... (Nearest integer)
Atomic mass of [Cu] = 63.5 U

Rough Work

MATHEMATICS **Max. Marks : 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. The value of $\left(1+\tan\frac{\pi}{14}\right)\left(1+\tan\frac{5\pi}{28}\right)$ is:

A) 1 B) 2 C) 3 D) 4

62. If α is a root of $25\cos^2\theta+5\cos\theta-12=0$, $\frac{\pi}{2}<\alpha<\pi$; then $\sin 2\alpha$ is equal to:

A) $\frac{24}{25}$ B) $-\frac{24}{25}$

C) $\frac{13}{18}$ D) $-\frac{13}{18}$

63. The value of $\cos\frac{\pi}{11}+\cos\frac{3\pi}{11}+\cos\frac{5\pi}{11}+\cos\frac{7\pi}{11}+\cos\frac{9\pi}{11}$ is:

A) 0 B) $\frac{1}{2}$

C) $-\frac{1}{2}$ D) None

64. If $A+B+C=\pi$ then $\cos 2A+\cos 2B+\cos 2C$ equals:

A) $1+4\cos A\cos B\cos C$ B) $1+4\sin A\sin B\sin C$

C) $-1-4\cos A\cos B\cos C$ D) $-1-4\sin A\sin B\sin C$

65. If $\frac{x}{\cos\theta}=\frac{y}{\cos\left(\theta-\frac{2\pi}{3}\right)}=\frac{z}{\cos\left(\theta+\frac{2\pi}{3}\right)}$ then value of $x+y+z$ is:

A) 0 B) $\frac{1}{2}$

C) $-\frac{1}{2}$ D) None

66. The range of $4\cos x+3\sin x-2$ is:

A) $[-5,5]$ B) $(-5,5)$

C) $[-7,3]$ D) $(-7,3)$

67. The minimum value of $\left|3\sin^2\theta-2\right|$ is

A) 5 B) 2 C) 1 D) 0

Rough Work

68. The sum to 10 terms of the series
$\frac{1}{1+1^2+1^4}+\frac{2}{1+2^2+2^4}+\frac{3}{1+3^2+3^4}+........$ is
A) $\frac{56}{111}$ B) $\frac{58}{111}$
C) $\frac{55}{111}$ D) $\frac{59}{111}$

69. The maximum sum of the series $100+98+96+.........$ is
A) 2500 B) 2550
C) 2050 D) 2555

70. If $\frac{x+y}{2}$, y, $\frac{y+z}{2}$ are in H.P., then x, y, z are in
A) A.P. B) G.P.
C) H.P. D) None

71. The number of real value (s) of x for which $\log 2, \log\left(2^x-1\right)$ and $\log\left(2^x+3\right)$ are in A.P. is
A) 1 B) 2
C) 3 D) 4

72. The minimum value of $1+2\sin x+3\cos^2 x$ is:
A) 3 B) $\frac{13}{3}$ C) – 1 D) – 3

73. $\frac{\sin 2A}{1+\cos 2A}$ equals
A) $\cot A$ B) $\tan A$
C) $\sin A$ D) $\cos A$

74. If $0<\theta<\pi$, then $\sin\theta$ lies in the interval:
A) $[0,1]$ B) $[-1,1]$
C) $(0,1)$ D) $(0,1]$

75. The value of $2\sin\frac{\pi}{8}.\sin\frac{2\pi}{8}.\sin\frac{3\pi}{8}.\sin\frac{5\pi}{8}.\sin\frac{6\pi}{8}.\sin\frac{7\pi}{8}$ is:
A) $\frac{1}{4\sqrt{2}}$ B) $\frac{1}{4}$ C) $\frac{1}{8}$ D) $\frac{1}{8\sqrt{2}}$

76. Let $f_k(x)=\frac{1}{k}\left(\sin^k x+\cos^k x\right)$ for $k=1,2,3,....$ then for all $x\in R$, the value of $f_4(x)-f_6(x)$ is equal to:
A) $\frac{1}{12}$ B) $\frac{1}{4}$ C) $\frac{-1}{12}$ D) $\frac{5}{12}$

Rough Work

77. Let α, β be such that $\pi < \alpha - \beta < 3\pi$.

If $\sin\alpha + \sin\beta = -\frac{21}{65}$ and $\cos\alpha + \cos\beta = -\frac{27}{65}$, then the value of $\cos\left(\frac{\alpha-\beta}{2}\right)$ is:

A) $\frac{-6}{65}$ B) $\frac{3}{\sqrt{130}}$

C) $\frac{6}{65}$ D) $\frac{-3}{\sqrt{130}}$

78. Which of the following is true?

A) $\sin 4 > \sin 4^{o}$ B) $\tan 3 > \tan 3^{o}$

C) $\cos 2 > \cos 2^{o}$ D) None

79. If $3\sin\alpha = 5\sin\beta$ then the value of $\frac{\tan\left(\frac{\alpha+\beta}{2}\right)}{\tan\left(\frac{\alpha-\beta}{2}\right)}$ is:

A) 2 B) 4

C) 1 D) None

80. If $\theta = 15^{o}$ then $\frac{1-\tan^2\theta}{1+\tan^2\theta}$ equals:

A) $\frac{1}{2}$ B) $\frac{1}{\sqrt{2}}$

C) $\frac{\sqrt{3}}{2}$ D) None

Rough Work

SECTION-II
(Integer Value Answer Type)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals**. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:
Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

81. The value of $\frac{\tan 80^\circ - \tan 10^\circ}{\tan 70^\circ}$ is ______.

82. The value of $\sqrt{3}\operatorname{cosec} 20^\circ - \sec 20^\circ$ is ______.

83. The value of $\sqrt{2+\sqrt{2+\sqrt{2(1+\cos 8\theta)}}} = k\cos\theta$, then k is ______ where $0<\theta<\frac{\pi}{8}$.

84. If $A = 2\sin^2\theta - \cos 2\theta$, then A lies in the interval $[a,b]$ then value of $|a-b|$ is _______.

85. If the value of $\cos^2 10^\circ + \cos^2 50^\circ + \cos^2 70^\circ$ is 'k' then $4k$ equals _______.

86. If $\cos 6^\circ . \sin 24^\circ . \cos 72^\circ = \frac{1}{\alpha}$ then α is _______.

87. If $\tan\frac{\pi}{9}, x, \tan\frac{7\pi}{18}$ are are in arithmetic progression and $\tan\frac{\pi}{9}, y, \tan\frac{5\pi}{18}$ are also in arithmetic progression then $|x-2y|$ is equal to _______.

88. The value of $(0.16)^{\log_{2.5}\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\ldots\text{to }\infty\right)}$ is _______.

89. The value of $\cos 1^\circ . \cos 2^\circ . \cos 3^\circ ....\cos 179^\circ$ is ________.

90. If $A+B=\frac{\pi}{4}$ then $\tan A + \tan B + \tan A \tan B$ equals _______.

*****

Subject	Physics	Chemistry	Mathematics
Setter Branch	VARANASI-NSPIRA-CC		
SME Name	RAMKUMAR KUSHWAHA	S.P. GUPTA	PRASHANT KUMAR
Phone Number	7578966757	9507184444	8229816866

Rough Work